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The flow of powder into simple and stepped dies

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Abstract

An experimental study of die filling into constrained cavities of simple and complex geometries is presented. A model shoe system for the study of powder flow in air and vacuum has been developed. Transparent dies and shoes have been used, which allow the flow and rearrangement of the powder to be observed using a high-speed video system. Both qualitative and quantitative estimations of the influence of powder characteristics and shoe kinematics on the die filling process have been made. Measurements of filling rate and filling ratio for various shoe and die geometries are presented. The influence of airflow is revealed by comparison of experiments conducted in air and in vacuum. Possible positive and negative contributions of the entrained air during die filling have been addressed. Results are presented in the context of the Beverloo equation, which was originally developed to characterize hopper flow. The concept of a critical velocity is introduced, below which complete filling of a standard die is achieved. This provides a measure of the flowability of a powder and can be used to gauge the influence of airflow and pressure on the filling process.
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1. Introduction

The physical and mechanical properties of powder metallurgical (P/M) components are closely related to their final density. Minimizing the density gradients is an important consideration when high and consistent mechanical performance is required. The final density is determined by the press-and-sinter process parameters and by the material characteristics and filling conditions. The density distribution of the material in the as-poured condition has effects that are propagated throughout the subsequent P/M processes. Understanding and quantifying the main factors that influence the fill density could be a platform for ''tailoring'' the initial density in the die.

Modelling flow in the context of powder metallurgical processes is still very much in its infancy. There is an extensive literature on bulk powder flows, with detailed studies on the discharge of powders from hoppers [1,2] and avalanches down inclined slopes [3,4]. There have, however, been limited studies on the type of constrained flows that occur during die filling. Previous studies of die filling have mainly concentrated on the factors which influence segregation and on characterizing the packing of powder inside a die after the filling process. For instance, Lawrence and Beddow [5,6] investigated powder segregation during die filling for the flow of a mixture of two different sized (fine and coarse) powders from a stationary hopper. They proposed that segregation occurs as a result of finer powder filtering down through the moving powder bed. They showed that segregation can be reduced by either decreasing the fill time or increasing the drop height. When a vibrating die is used, vibration reduces horizontal segregation since the fine powder tends to level off, while the effect of vibration on vertical segregation depends on the stability of the powder bed in the die. Demetry et al. [7] used tactile sensors to explore the uniformity of density distribution in a die. The density uniformity was inferred from the measurement of pressure distribution at the base of the die. They showed that a pouring device that delivers the powder directly above the die opening rather than from the side of the die leads to a more uniform pressure distribution. The influence of die width on filling density was investigated by Bocchini [8], who showed that the filling density decreases as the die opening decreases. Rice and Tengzelius [9] evaluated how the packing density depends on the details of the filling system. They showed that the filling density

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increases as the shoe speed decreases. Also, vibration generally increases the fill density, except that vibrating at too high a frequency leads to fluidisation and hence to low density in certain dies. It is evident that the extent of any segregation and the filling density depend upon the details of the flow process. The full identification of the microscopic and macroscopic features of the flow process and how this is influenced by the kinematics of the filling procedure, the die geometry and the powder characteristics would represent an important step towards the development of a suitable model of die filling. The present study presents both qualitative and quantitative results, which attempt to achieve this objective.

The details of the filling process and the way in which the powder finally packs within the die involve a complex interaction between the filling system and the cavity which is being filled. The most commonly used filling device in the P/M industry consists of a rectangular-sectioned box having a forward motion of constant velocity over the die cavity, a number of short 'shakes' followed by reverse motion over the cavity. The shoe velocity currently employed in industry is generally in the range of 50–450 mms^{-1}, with typical values around 100 mms^{-1}. The shakes consist of forward-and-reverse motions of higher velocities (e.g. 250 mms^{-1}) over very short distances (e.g. 15 mm). Before considering this type of commercial process, it is instructive to first consider some simpler flow conditions. There is an extensive literature on hopper flow (this is perhaps the simplest example of a delivery system) and sand pile formation on a flat uniform surface (which is the simplest example of a collection system for flowing powder). These studies are reviewed in Section 3 and the major observations and relationships that are relevant to die filling are presented.

In order to examine the influence of the motion of the powder delivery system on the die filling process, a model shoe filling system has been designed. This is fully described in Section 4. We then consider the interaction between the filling system and die cavity. The simplest system of this type consists of flow from a stationary shoe or

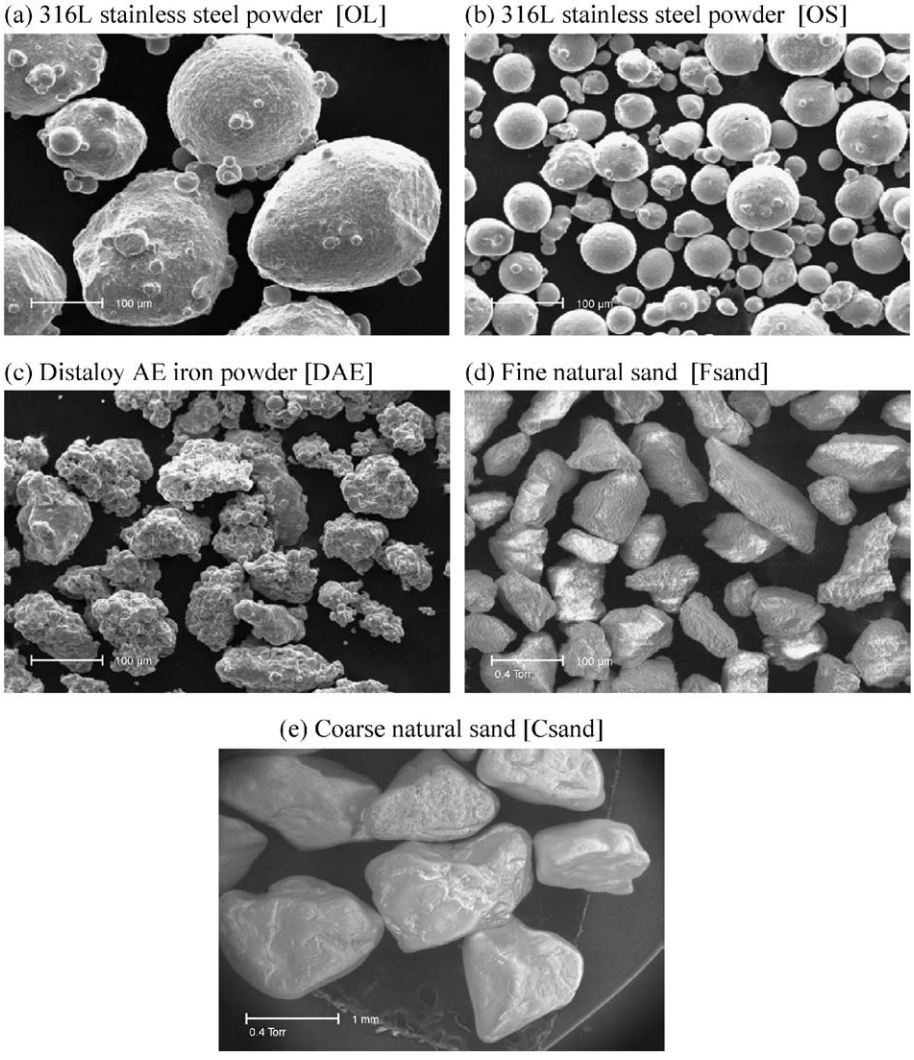


Fig. 1. SEM images of various powders used.

hopper into a simple uniform cavity of rectangular cross-section, which is presented in Section 4.1. Experiments involving the flow of powders into simple shaped cavities from a moving shoe are presented in Section 4.2 and the results of experimental studies using dies with features, such as steps, are described in Section 4.3. In each of these studies, a range of powders with different physical characteristics have been used, as described in Section 2. The influence of shoe motion is evaluated from tests conducted using different shoe velocity profiles. The interaction between air and particle flow is evaluated by comparing the results of experiments conducted in air and vacuum and by employing different shoe designs. A high-speed video system has been used to determine the detailed arrangement of powder in the moving shoe, the flow of powder into the die and how it packs to form the starting condition for a green body. These studies allow those features which most critically influence the flow process and the way in which the powder packs into a die to be identified, thus providing the necessary background information for the construction of constitutive models for powder flow and providing guidelines for the design of powder delivery systems.

2. Powder characteristics

In order to gain a fuller understanding of the nature of the flow process and to investigate the influence of the full range of factors identified above, a number of different flow experiments were conducted on a range of powders. Five different powders were employed in the experiments described in this paper. SEM micrographs of the different powders are shown in Fig. 1 and some relevant geometric and material parameters are listed in Table 1 together with the designation of the powders. Three of the powders tested were steels, with roughly the same material density. The dimensions of these metal powders, as given by the producer, were in the following size range: Distaloy AE (DAE), 25% less than 45 μm, 75% in the range 45–150 μm; Gas atomised stainless steel powder of small grain size (OS), 20–106 μm; Gas atomised stainless steel powder of large grain size (OL), 106–250 μm. The powder morphology was evaluated by image analysis. For each material, between five and seven SEM images were digitised and processed using ImagePro® (ex-Media Cibernetics). The 2-D projected area, A, and perimeter, P, were measured for each particle, and the overall shape factor, $S=4\pi A/P^2$, was calculated after the statistical processing of data (see Table 1). The definition of the shape factor is such that $S=1$ for spherical particles, which is the maximum possible value of S. The smaller the value of S, the more irregular the particle shape. This parameter should be examined in conjunction with the images shown in Fig. 1. The stainless steel powders are smooth and roughly spherical, while the Distaloy powder is highly irregular. The sands are roughly equiaxed with faceted boundaries. Experiments employing the three steel powders allow the influence of particle size and shape to be determined, while keeping the material density constant. Comparing these experiments with those for sands allows the influence of material density to be investigated.

Table 1
Characteristics of the materials employed

Material	OL, 316L powder	OS, 316L powder	DAE, prealloyed iron powder, Distaloy AE™	Fine sand, natural sand	Coarse sand, natural sand
Average diameter (D_{50}) (μm)	150	55	90	120	1384
Shape factor ($S=4\pi A/P^2$)	0.81	0.84	0.73	0.72	0.71
Apparent density, (Mgm^{-3})	4.41	4.58	3.15	1.25	1.51

3. Hopper flow and sand pile formation

There have been extensive studies on hopper flow and sand pile formation. These processes are simpler than the die filling process which forms the central theme of this paper. It proves useful, however, to first consider these simple situations and to identify the major characteristics of the processes and how these are influenced by the powder characteristics and geometric parameters. In this section, we identify the major results that are relevant to the study of die filling.

3.1. Hopper flow

Hopper flow has been extensively studied due to its technological importance in powder and granular material handling industries. In general, the flow pattern is dependent on the hopper geometry, friction between the particles and the hopper wall, as well as on the material internal friction. Previous work has been undertaken on modelling the influence of hopper angle and wall friction on the stress and velocity distribution [1,2,10].

The main concern in material processing industries is the mass flow rate, which characterizes the flowability of the powder through a hopper. A higher flow rate is required for faster and more efficient processing. The simplest example of hopper flow is when powder flows under gravity through a hopper with the top and bottom of the hopper exposed to atmosphere, so that the pressure at these locations is identical. This is referred to as gravity flow. During the discharge process, the powder in the hopper can be classified into two types of region: a central flowing region and stagnant regions close to the walls, as shown in Fig. 2. The size of the stagnant region depends on the particle size and the hopper angle (which is defined as the angle of the walls with respect to the horizontal surface). The stagnant zone decreases, even disappears, as the hopper angle increases. This generally occurs for hopper angles greater than about

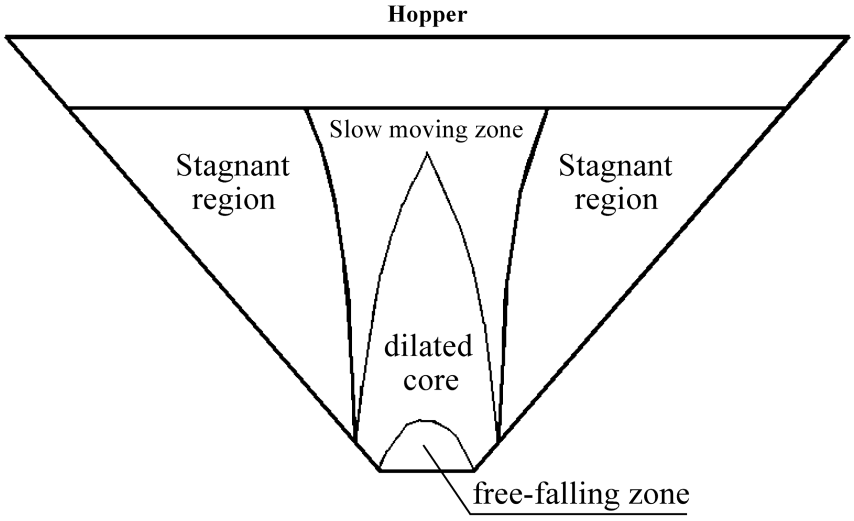


Fig. 2. Diagram of flow regions for powders discharging from a wedge-shaped hopper.

45°. Core flow, i.e. when the stagnant zones are present, is more relevant to shoe delivery systems and we concentrate on this situation in the remainder of this paper.

In the central flowing region, the particles at the bottom fall freely under gravity once they detach from the powder mass. This zone is normally referred to as a free-falling zone [11]. Above this zone, there is a core of fast moving material. It is observed from hopper flow experiments that the material in this core slightly dilates during the flow process (denser packing can be observed in the stagnant zone). The difference between velocities of the free-falling particles and those of the material in the fast moving dilated core implies that an arch is formed at the top of the free-falling zone, which resists the flow of material in the fast moving dilated core. With powder flowing in the central region, a significant depression in the top surface occurs. The depression grows until the slope exceeds a certain angle that is believed to be related to the angle of repose, whereupon the material cascades down the surface towards the centre. Detailed observations of the free-falling and fast moving dilated zone indicate that the velocity of the particles is greater at the centre compared to the edge of the flow. Under these conditions, the flow rate for coarse grained materials is well described by the Beverloo equation [1,2,11–14]:

$$M = C\rho_0\sqrt{g(D_H - 1.4d)}A^* \quad (1)$$

where ρ_0 is the bulk density of loosely packed powder, g is the acceleration due to gravity, D_H is the hydraulic diameter of the hopper opening (four times the cross-sectional area divided by the perimeter), d is the particle diameter and A^* is the effective discharge area (actual cross-sectional area minus $1.4 \times d \times$ perimeter) and C is a dimensionless constant with a value of 0.73 for most materials [2]. Eq. (1) is applicable only for powders with an average grain size larger than 0.5 mm and for powder heights greater than $2D_H$. The flow rate reduces with decreasing powder height below $2D_H$. For smaller particles, the effect of interstitial pressure gradients, arising from the expansion of the self-entrained gas as the powder mass dilates, becomes significant [15], especially for fine powders [1,2]. A modified Beverloo equation for the solid flow rate has been suggested by Crewdson et al. [15], which takes account of the influence of this pressure gradient:

$$M = M_0\left(1 + \frac{1}{g\rho_o}\left.\frac{dp}{dr}\right|_{r_0}\right)^{1/2} \quad (2)$$

where $(dp/dr)|_{r_0}$ is the pressure gradient at the orifice and M_0 is the mass flow rate predicted by Eq. (1). When powders of small grain size flow through a hopper, the percolation of the entrained air induces a pressure less than atmospheric in the interstitial fluid immediately above the orifice as the powder mass dilates. The material passing through the orifice then experiences an adverse (negative) pressure gradient [13], which inhibits the flow process. Verghese and Nedderman [16] have demonstrated that the magnitude of this adverse pressure gradient increases with decreasing particle size, resulting in a decrease of the mass flow rate. From a series of experiments on fine and coarse sands with particle sizes in the range 150–2500 μm, they found that the flow rate is given by

$$M = M_0\left(1 + \frac{\lambda}{d^2}\right)^{1/2} \quad (3)$$

where $\lambda = -1.46 \times 10^{-8}$. This result implies that the mass flow rate is zero for particle sizes less than 120 μm. Hopper flow experiments using the powders described in Table 1 produce measurable flow rates, as will be evident in subsequent sections. This result cannot therefore be extrapolated beyond the conditions employed in the experiments,

but it clearly demonstrates the influence of particle size on the pressure gradient and flow rate.

It is clear from Eq. (2) that the mass flow rate can be increased by properly controlling the pressure gradient in the hopper, instead of allowing the pressure gradients generated by the gravity flow itself. This is called pressurised hopper flow, in which air is deliberately supplied within the hopper to impose a pressure gradient. Pressurised hopper flows can be divided into two groups: positive flow and negative flow [17]. Positive flow is when the interpore pressure is greater than atmospheric. This is achieved by aerating the powder in the hopper. Ferrari and Poletto [18] observed a factor of 10 increase in the flow rate of fine powders in the range 60–150 μm by aerating to a fluidised state. Negative pressure flow is when high pressure is created immediately below the orifice. In steep angled hoppers, this can either prevent arches forming above the orifice or move the arch further up the hopper, giving a larger effective discharge area. Under these conditions, a faster flow rate can be achieved compared to pressureless conditions [17]. In shallow hoppers, however, a negative pressure always slows down the flow rate, i.e. when core flow occurs an external pressure reduces the flow rate, as predicted by Eq. (2).

3.2. Pile formation

There has been significant research interest in recent years on the formation of sand piles. Much of this has focused on the nature of the pressure distribution developed at the base of the pile [19,20]. A full evaluation of this is beyond the scope of the current paper. Here, we simply examine some characteristics of the formation process, which can provide us with useful background information for the more complex problem of die cavity filling considered in this paper.

Preliminary experiments were conducted to construct piles by axisymmetric flow from a conical hopper of height 60 mm and 60° angle, with a 12-mm diameter aperture. Piles were created using various pouring heights, measured from the base of the pile (50–300 mm), using two materials (fine sand and DAE, see Table 1). The hopper was filled to the top in both cases, so that the same volume of materials was employed. The shape of the pile and the details of the formation process are a function of the pouring height. At high heights, the falling stream causes the formation of a crater on the growing pile surface. When the stream hits the bottom of the crater, some of the particles collide with the incoming stream and slide in waves on the pile surface. There are particles with sufficient kinetic energy to cascade over the pile surface, forming a cloud around the upper part of the pile. The large particles roll on the pile surface and splash outside the main body forming a layer that surrounds the pile base, like a crown. This crown is more evident during the initial stage of pile formation. Small pouring heights result in a steadier flow. There is less kinetic energy as the powder impacts with the pile. There is therefore limited disturbance of the underlying powder particles and no crater is observed. The pile develops by the successive build up of layers which form by the localised flow of powder over the surface of the pile, similar to that described above. A sharp apex then develops as the pile grows. There is a noticeable change of slope of the pile from base to tip. The decrease of angle of repose is believed to be related to a slight decrease of bulk density and internal friction towards the freshly formed regions near the peak of the pile.

Even in these simple flow situations, there is a rich range of phenomena, which can be changed by changing the characteristics of the powder and geometric features such as hopper angle and pouring height. We build on this understanding in the following section, where we gradually increase the complexity of the situations considered.

4. Flow of powder into a die

The above experiments and discussion illustrate the important role that both the delivery and collection systems play in determining the final state of the deposited material. It is therefore important to reproduce, as closely as possible, the physical conditions that exist in practical die filling situations. A model shoe system was therefore designed to allow a more realistic simulation of commercial die filling processes (see Fig. 3). The system consists of a pneumatic driving unit, a positioning controller and a displacement transducer. The main system parameters are the following: shoe travel distance 220 mm; speed range 0.033–1 ms^{-1}; acceleration range 0.1–100 ms^{-2}. Various velocity profiles can be achieved, which allows the influence of the details of the shoe motion to be evaluated. Transparent dies of different geometries were employed to study the influence of die size and shape. The entire system is located in a vacuum chamber with glass walls, within which a vacuum of less than 1 Torr can be achieved. By conducting tests in air and vacuum, the influence of airflow and air pressure on the filling process can be determined. A high-speed video system (type NAC500)

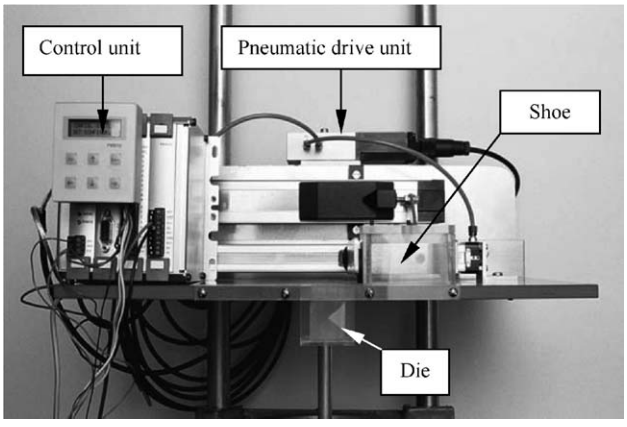


Fig. 3. The experimental set-up for die filling study.

was used for qualitative observations of die filling processes. A typical shoe velocity profile is shown in Fig. 4. This profile is characterized by the initial acceleration of the shoe, a_i, and the steady state velocity, v_{shoe}. In the experiments conducted to date, the initial acceleration was varied over the range of 0 to 100 ms^{-2} and the steady state velocity was varied over the range of 0 to 1 ms^{-1}.

In the following sections, we examine the influence of shoe motion and die geometry on the kinematics of the die filling process. We also examine the influence of powder characteristics by conducting experiments on the full range of powders described in Section 2, Table 1 and Fig. 1. In Sections 4.1 and 4.2, we consider the filling of simple rectangular cavities and examine how the kinematics of the shoe motion influences the filling process. In Section 4.1, we describe a series of experiments in which the shoe is stationary and the powder flows by gravity into a simple straight cavity of rectangular cross-section. This can be viewed as a constrained version of hopper flow and sand pile formation considered in Section 3. This is the simplest condition in which there is an interaction between the delivery system and the die. We then gradually increase the complexity of the situation being considered and in the process develop our understanding of the physical mechanisms which determine how a die is filled. In Section 4.2, we examine the influence of the shoe kinematics on the flow process by employing shoe velocity profiles of the form illustrated in Fig. 4 and systematically varying the initial acceleration and steady state velocity over the die. The influence of die shape is examined in Section 4.3. A series of experiments are described involving stepped rectangular dies. For the stepped die, two different cases have been considered by creating a narrow section either upstream or downstream of the direction of shoe motion. The influences of shoe speed, powder characteristics and airflow on die filling processes are discussed. In all the experiments described here, a prescribed volume of material was placed in the shoe at the start of an experiment. In practical filling systems, the shoe is generally continually filled through a hopper feed system. The results presented here are influenced by the fact that the mass of powder in the shoe gradually decreases as the die is filled, but these experiments allow us to focus on the interaction of the shoe with the die and the results are not further complicated by interactions with a hopper fill system.

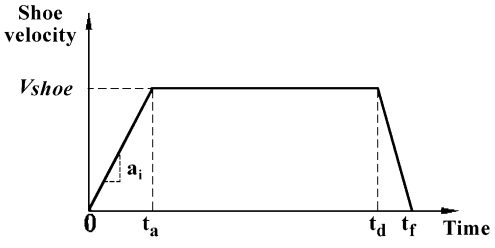


Fig. 4. The shoe velocity profile: ($0 - t_a$) acceleration period, ($t_a - t_d$) steady velocity period, ($t_d - t_f$) deceleration period.

In the discussion of hopper flow presented in Section 3, it is evident that air pressure gradients generated as the powder mass dilates at the orifice of a hopper strongly influence the flow rate. In the following subsections, it will become evident that there are a number of additional effects of air pressure and flow which need to be taken into account in order to understand and model the filling process. The full range of tests described above has therefore also been conducted in vacuum. By comparing the results of experiments in air and vacuum, the various effects of air on the filling process can be evaluated.

4.1. Flow from a stationary shoe

Initially, we consider the situation where the shoe is stationary and is located directly over a simple rectangular straight cavity. A thin sheet of plastic initially covers the cavity. As this is removed, the powder flows freely into the cavity. We start by describing the flow into a cavity 23 mm wide, 25 mm long and 38 mm deep, which we refer to as a wide cavity. At the beginning of the filling process, the powder mass is stationary and the front of the falling powder is level with the top of the die. However, with the evolution of filling, the front of the falling powder appears to form a concave shape. This is because the flow velocity in the centre is higher than at either side, as observed by Tuzun et al. [11] for hopper flow. The flow behaviour inside the shoe is identical to core flow from a hopper as discussed in the previous section and illustrated in Fig. 2. In this constrained configuration, it can be difficult for the air to escape from the die as it is being filled. As it tries to escape, turbulence is created in the powder stream. This effect is most pronounced for the fine (OS and DAE) and low density (fine sand) particles. Once the air reaches the top of the die, there are two major routes by which it can escape from the die: either by permeating through the powder mass above the die; or by flowing through the small gaps created between the shoe and die. Evidence for this latter process can be seen in the high-speed videos as the expulsion of powder in the front and back of the shoe. This is most pronounced for the fine, light particles (DAE, OS and Fine sand). Air can more readily permeate through the coarser particles and gravitational forces always exceed the drag forces exerted by the expelled air. Thus, there is limited turbulence and expulsion of particles for the OL and coarse sand powders. These observations can be compared with those made on tests conducted in vacuum. There is now no turbulence in the powder stream and the particles simply fall vertically after they detach from the powder mass in the shoe. As before, a concave-shaped front is formed shortly after the flow starts due to the variation in flow velocity across the opening [11]. Consequently, the powder builds up as a heap with an apex at the centre.

Similar experiments have been conducted using a die with dimensions 5 mm wide, 25 mm long and 38 mm deep, which we refer to as a narrow die. The details of the filling process

are similar to those described above for the wide cavity. The only difference is that a smaller proportion of the powder originally in the shoe flows into the die. Whereas the level of the powder in the shoe immediately above the wide die after filling is almost coincident with the top surface of the die, the height of powder above the narrow die is several times the hydraulic diameter throughout the filling process.

The Beverloo equation presented in Section 3 does not strictly apply to the conditions considered here. This equation relates to steady state flow conditions from a hopper, while a steady state is never achieved during these die filling experiments. It is instructive, however, to examine these experiments in the context of this equation and the discussion of Section 3. The average flow rate and the implied constant C obtained by fitting the raw flow data to the generic Beverloo relationship of Eq. (1) are given in Table 2 for four of the materials employed in this study. Here the filling rate $\bar{M}$ is defined as

$$\bar{M} = \frac{m}{t} \tag{4}$$

where m is the final mass of powder in the cavity, and t is the duration of filling and is defined as the time interval between the start of flow and the cavity being completely filled. In the present study, the filling process was recorded using a high-speed video system (500 fps) and the filling duration was obtained by counting the frames. The typical filling duration is ca. 0.3–0.5 s, so that the error in determining the filling duration t is around 0.4–0.7%. The powder in the cavity was weighted using scales with a precision of 0.01 g. Reproducibility of the tests was verified by repeating the experiments five times. The standard deviation is normally less than 10%.

It is evident from Table 2 that filling rates in vacuum are always faster than those in air. The difference ε between the experiments conducted in air and vacuum are also given in Table 2, where

$$\varepsilon = \frac{\bar{M}_v - \bar{M}_a}{\bar{M}_v} \tag{5}$$

with $\bar{M}_v$ and $\bar{M}_a$ representing the filling rates in vacuum and air, respectively. This difference is most pronounced for the flow of fine materials into the wide cavity where more than a factor of 2 increase in flow rate is observed in vacuum. The effect is less pronounced for the narrow cavity, probably because of the reduced level of turbulence in this die and the increased fill time. The general trend observed here is consistent with the concepts that underpin the Beverloo equation. The finer the particles, the lower the effective flow rate in air (characterized in terms of the parameter C of Eq. (1)). In the modified Beverloo equation of Eq. (2), this effect is explained in terms of the effect of pore pressure gradients. In addition to the effect of pressure gradients arising from dilation of the powder mass, we must add the effect of the build up of air pressure in the die as it is filled with powder. This depends on how fast air is able to escape from the die compared to the time it takes to fill the die. According to the Beverloo equation, the time to fill the die is inversely proportional to the square root of the hydraulic radius, thus a wide die is expected to be filled faster than a narrow die. This gives less time for air to escape, either by percolating through the powder mass or leaking through the gap between the shoe and die. Thus, a higher air pressure is built up in the wide die, resulting in a higher pore pressure gradient and a slower than expected flow rate, i.e. a smaller value of C (see Table 3).

Table 2
Comparisons of the filling rate (gs^{-1}) in air and in vacuum for various powders flowing into a cavity

Powder	Narrow cavity			Wide cavity		
	In vacuum	In air	ε (%)	In vacuum	In air	ε (%)
Coarse sand	12.82	11.87	7.35	139.08	128.72	7.44
OL	75.77	58.63	22.62	479.04	381.99	20.26
OS	61.57	47.57	22.74	391.00	226.22	42.14
DAE	57.77	35.44	38.65	371.60	151.55	59.22

The effect of air pressure and pressure gradients is removed by conducting tests in vacuum, so according to Eqs. (1) and (2), we would expect the same value of C for all materials, particularly for the same test geometry and the OS and OL powders, where the only difference is the mean particle size. Smaller values of C are however obtained for the smaller particles in vacuum, thus, air pressure gradients only account for part of the difference in the behaviour between these materials. For finer powders (say OS), surface interactions of the particles are stronger, which can result in arching and inhibition of the flow of powder into the cavity. Nevertheless, it is interesting to note that the values of C for OL and DAE, which have larger particles, obtained by filling the narrow cavity in vacuum are comparable to the classical Beverloo constant observed during hopper flow. The smaller values of C for the wide cavity compared to the values for the corresponding test using the narrow cavity are a result of the lower average height of the powder during the filling process. In these cases, the height of the powder inside the shoe is about 40 mm, which is less than the condition for the Beverloo equation to be applicable ($H>2D_H$). Also, for coarse sand, values of C larger than 10 are obtained when filling the narrow cavity, which is significantly larger than the values obtained for the other powders. This is attributed to the fact that the size of the opening is too small ($<6d$) and is beyond the limit of applicability of the Beverloo equation [2].

4.2. Flow from a moving shoe

In this section, we consider the influence of shoe motion on the mechanism of the filling process. Initially, the shoe was positioned away from the cavity. As with the experiments described above, a fixed volume of material was placed in the shoe. The shoe moves from rest and follows

Table 3
Comparisons of parameter C in air and in vacuum for various powders

Powder	Narrow cavity		Wide cavity	
	In vacuum	In air	In vacuum	In air
Coarse sand	11.1584	10.3271	0.7186	0.6652
OL	0.6944	0.5374	0.5391	0.4300
OS	0.5015	0.3874	0.4118	0.2382
DAE	0.7042	0.4320	0.5750	0.2345

the prescribed velocity profile of Fig. 4. A typical die filling experiment is shown in Fig. 5, which shows a sequence of images taken from an experiment in which fine sand is delivered into a wide cavity. The initial acceleration of the shoe was 1 ms^{-2} and it reached a steady velocity v_{shoe} of 300 mms^{-1}. This figure illustrates a number of important features of the die filling process. Due to the effect of inertia when the shoe accelerates from rest and frictional sliding of the powder mass on the base, the powder moves towards the back of the shoe. Since the powder particles near the top free surface can move more easily compared to those near the bottom of the shoe, a nose-shaped pattern (or a depression at the front of the shoe) is generally developed as the powder is packed towards the back wall during the initial acceleration of the shoe. A gap between the front wall and the powder is created if the initial acceleration of the shoe is sufficiently high or the powder bed in the shoe is relatively thin. This causes some densification of the powder since some voids inside the powder may collapse under the intense motion of the particles, leading to increased cohesion of the powder. When the shoe moves across the die, the powder flows into the cavity under gravity (Fig. 5a). Two kinds of flow occur successively (see Figs. 5b and 6). The first of these, which we refer to as "nose flow", can be described as a landslide of the front region of the powder when the nose is destabilized as it moves over the die cavity. Since this involves a free surface, it is very easy for the first layer of grains to detach and to fall freely into the die. At

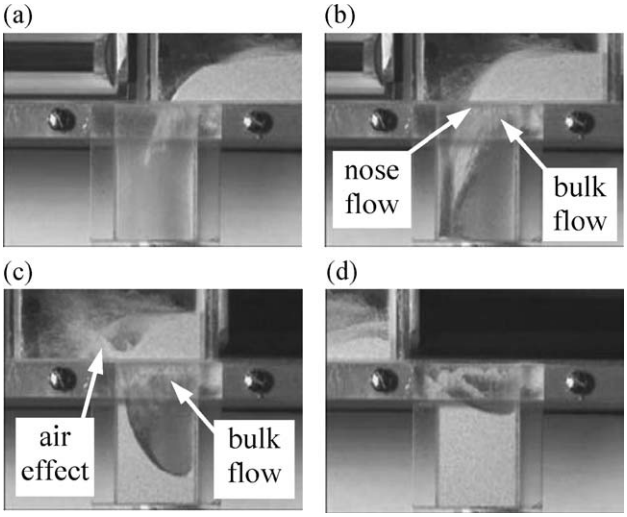


Fig. 5. Typical die filling process (fine sand, wide cavity, V_{shoe} = 300 mms^{-1}).

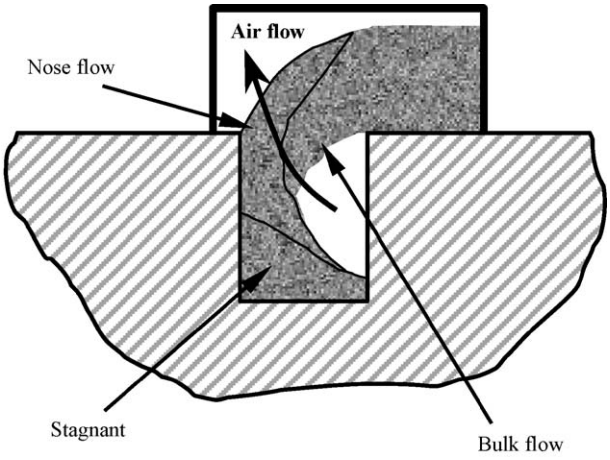


Fig. 6. Illustration of general features of die filling process.

this particular shoe speed, the stream of falling powder initially builds up in the leading corner. Since the powder is delivered by cascading down the surface of the nose, it primarily comes from the top of the shoe. As the filling proceeds further, the powder mass builds up in this corner and then flows back towards the trailing corner. As the tip of the nose translates across the die cavity, some powder detaches directly from the bottom surface of the powder that previously slid over the flat platform. We refer to this type of flow as "bulk flow". For the conditions employed in Fig. 5, material from the bulk fills the trailing corner of the die, as illustrated in Fig. 6. This is further illustrated in Fig. 7a, which shows the final powder distributions observed in an experiment in which two layers of coloured sand were initially deposited in the shoe. It is evident from this experiment that the top layer of sand has supplied more material into the die than the bottom layer, indicating that nose flow dominates the filling process. Bulk flow becomes dominant if the shoe moves so fast that the tip of the nose completely translates over the cavity before any powder flows into the die (such as when filling a narrow cavity or filling a wide cavity at a high shoe speed). This is illustrated in Fig. 7b, which shows the result of an experiment using two layers of coloured sand for a shoe velocity of 500 mms^{-1}. At this velocity, the cavity is only partially filled and all the powder is delivered by bulk flow from the bottom of the shoe.

Another important observation from the series of images shown in Fig. 5 is the influence of the entrapped air once the tip of the nose moves over the cavity. As the air pressure builds up in the die, the entrapped air tries to escape. The air escapes through the thinnest section of the flowing powder stream and for light, fine powder particles the stream of expelled gas carries some of the particles with it, as can be observed in Fig. 5c.

Experimental studies in which the wide and narrow dies have been filled in air and vacuum has been conducted employing shoe velocities over the range 0–400 mms^{-1} using the full range of powders described in Section 2. The initial acceleration is found to have only a small influence

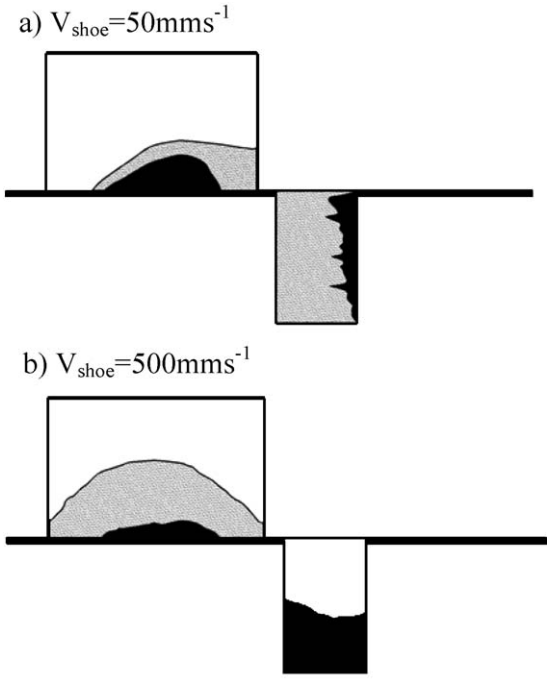


Fig. 7. View of a rectangular die (23 × 20 × 38 mm) after filling at different shoe speeds (sand, shoe motion from right to left, $a_i = 50$ mms^{-2}). The powder was initially deposited in the shoe to form two horizontal layers, with the darker layer at the bottom.

on the observed behaviour. In all the results reported here, an initial acceleration of 100 ms^{-2} was employed. The average flow rate obtained in these experiments is plotted in Fig. 8. As before, it is instructive to examine these results in the context of the Beverloo relationship of Eq. (1). The values of the implied constant C obtained by fitting the raw flow data to Eq. (1) are given in Fig. 9. In these and all subsequent figures, the solid bars represent the results in air, while the open bars represent the additional contribution to the quantity being measured from tests in vacuum.

There are a range of effects and phenomena which combine to produce the behaviour illustrated in Figs. 8 and 9. We start by considering flow into the narrow cavity. For this die, under all the conditions employed in this study, the powder is delivered into the cavity by bulk flow. The net flow rate and apparent value of C gradually decrease with increasing shoe velocity. There are two related effects which contribute to this. When the delivery system is stationary an arch forms above the opening of the cavity and a core of material is mobilised which feeds into the die. When the delivery system moves, the arch must move relative to the powder mass and new material must be mobilised to feed into the die. The effective arch will generally form above the top plane of the die. Thus, all the material which detaches from the arch will not enter the cavity. Also, during steady state flow from a stationary shoe, the particles build up speed as they are channelled towards the opening. When the shoe moves, there is less time for the newly mobilised material to achieve the steady state velocity of the fully developed flow, thus, the particles entering the cavity are likely to have a slower vertical velocity when the shoe is moving. Also, as the velocity increases, it can be more difficult for particles to detach from the bulk due to interlocking of the particles, which is enhanced by lateral compressive stresses generated by inertia effects.

Other effects come into play when considering the flow into the wide cavity. The most noticeable features of Figs. 8 and 9 are that there is a sudden decrease in the mass flow rate and apparent value of C as the shoe velocity is increased from zero to 100 mms^{-1} and the small difference between the results in vacuum and air at this higher velocity. At a shoe velocity of 100 mms^{-1}, nose flow dominates the filling process. As the shoe moves over the die, powder at the front of the shoe rapidly falls into the die. A space is then formed at the front of the shoe which grows as the shoe translates. There are both negative and positive features of

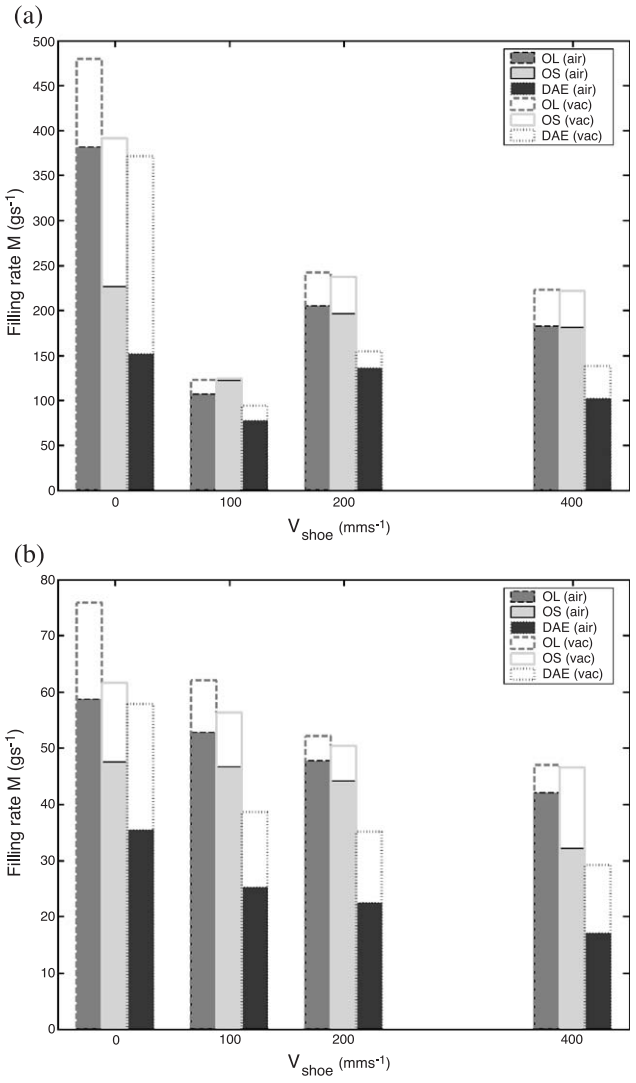


Fig. 8. Filling rates for various powders flowing into (a) a wide cavity and (b) a narrow cavity in air and in vacuum.

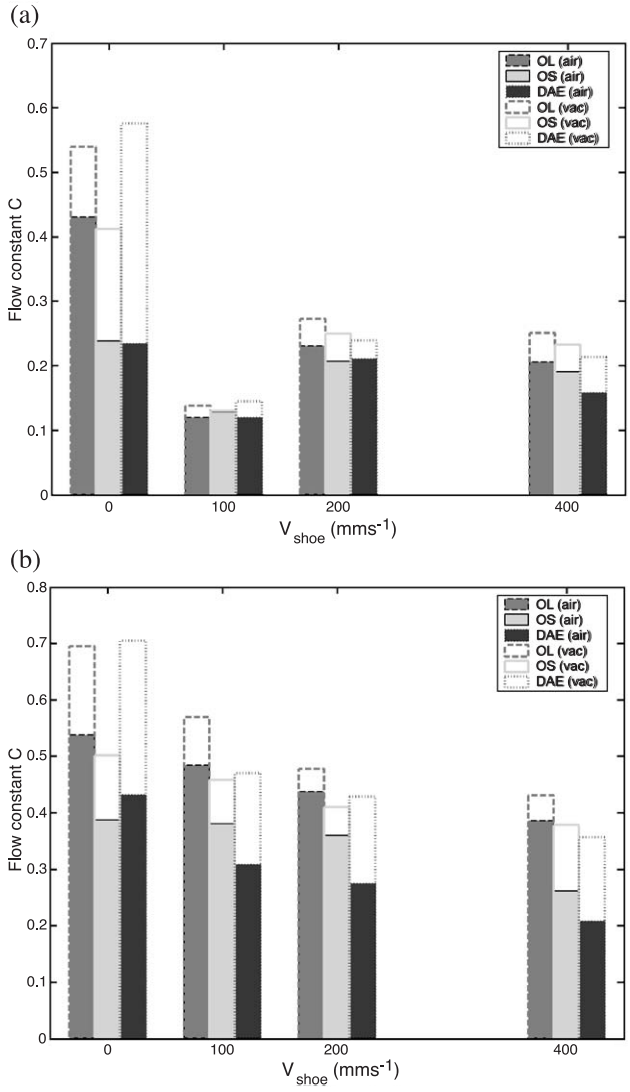


Fig. 9. Flow constant C for various powders flowing into (a) a wide cavity and (b) a narrow cavity in air and in vacuum.

this in terms of its influence on the mass flow rate. Firstly, the creation of this space means that powder is not delivered into the die across its full cross-section. The slower the translational velocity of the shoe, the smaller the effective area through which the material flows and the slower the mass flow rate into the die. This is counterbalanced by the fact that nose flow dominates the filling process; the cascade of powder over the surface of the nose is inherently a faster process than detachment from the bulk. Also, the creation of the nose provides an easy channel for air to escape, hence the small difference between the results in air and vacuum.

As the velocity is increased to 200 mms^{-1}, the tip of the nose translates faster across the cavity and powder is suspended in the shoe above the cavity. Powder is delivered into the die over a larger area by a combination of nose and bulk flow. When about 30% of the cavity is filled, the tip of the nose has translated across the cavity and pressure builds up in the die, which opposes the flow process. The net effect of this is that the average flow rate into the die is greater than that at 100 mms^{-1}, but there is now a noticeable difference between the results in air and vacuum. This arises from the increased contribution of bulk flow to the filling process and the developed pressure gradient as powder detaches form the bulk, which is further enhanced during the later stages of the filling process due to the additional pressure generated by the trapped air. Increasing the shoe velocity to 400 mms^{-1} results in an increase in the contribution from bulk flow and a reduction in the net flow rate into the die in both air and vacuum for all the powders tested. The reasons for this are similar to those described above to explain the decrease in filling rate for the narrow cavity.

It is interesting to note that at lower shoe speeds, in which nose flow dominates the filling process and there is a limited influence of air pressure, the powder can flow freely from the tip of the powder bed and the flow constant C has similar values for all the powders tested, i.e. $C \approx 0.14$ in air and vacuum. However, for the situations in which bulk flow dominates, the influence of cohesion becomes significant and the value of the flow constant C depends on the powder characteristics. This phenomenon becomes more significant for filling at higher speeds, particularly in vacuum. This is not surprising since the degree of increased cohesion due to interlocking of the powder particles becomes the dominant factor in vacuum because the effect of air pressure is eliminated. This effect is more pronounced for finer and more irregular powders (e.g. DAE).

At the highest velocities used in the testing programme, both the narrow and wide cavities are only partially filled after a single transversal of the shoe. If C in the Beverloo equation is constant and the packing density in the shoe after the filling process is simply the apparent density ρ_0, it is a relatively straightforward process to determine the critical velocity, v_c, at which complete filling occurs and the fraction of the die which is filled at higher velocities, δ. If the particle size d is much smaller than the characteristic dimensions of the opening then

$$v_c = C\sqrt{gD_H}L/h \tag{6}$$

and

$$\delta = v_c/v \tag{7}$$

where L is the length of the shoe and h is the height of the cavity being filled.

Fig. 10 shows the filling ratios obtained for the flow of DAE into a narrow cavity at various shoe speeds. For each data point in this figure, five tests were conducted and the error bars show the standard deviation. The fit of Eq. (7) to this data is also plotted in Fig. 10, where v_c was treated as a free parameter to provide the best fit to the data. It is evident that Eq. (7) provides a reasonable first approximation of the

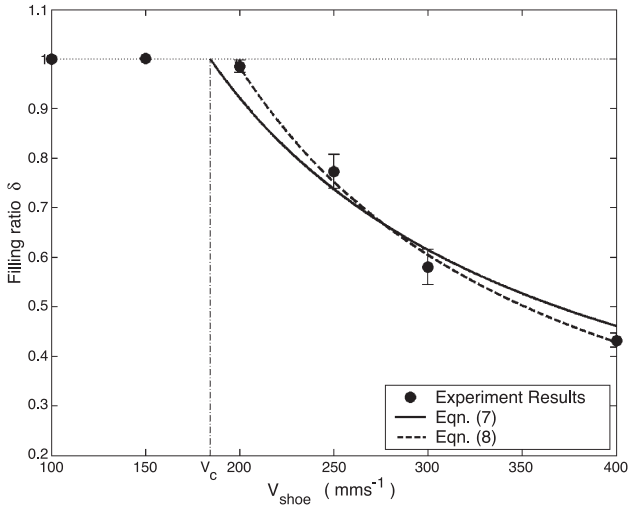


Fig. 10. Filling ratio for DAE flowing into a narrow cavity at various shoe speed.

degree of filling, but it underestimates the critical velocity and the filling ratio at low shoe speeds, but overestimates the filling ratio at high shoe speeds. This divergence between experimental data and Eq. (7) is not surprising because C decreases with increasing velocity, as indicated in Figs. 8 and 9.

A better fit of the experimental data is obtained using an empirical expression of the form

$$\delta = (v_c/v)^{6/5} \tag{8}$$

which is plotted as a dashed line in Fig. 10. This provides a more accurate fit of the critical shoe velocity and the filling ratio over the full range of shoe speeds. Eq. (8) implies that C is proportional to $v^{-1/5}$. This result is broadly consistent with the results of Figs. 8 and 9.

Either Eq. (7) or (8) can be used to guide the choice of shoe kinematics in a given situation. These equations can be used to determine the critical velocity in a simple manner since only two tests need to be conducted: one is at low shoe velocity where the cavity is completely filled; the other is at a high shoe speed where partial filling is obtained. From these two tests, the filling ratio can be determined and the critical velocity can then be obtained from Eq. (7). The critical velocity for various powders considered here have been determined using this approach and the results are given in Table 4. By definition, a higher value of the critical velocity indicates an easier flowing powder since the cavity can be completely filled at higher shoe speeds. It is clear that the critical velocity for filling in vacuum is always higher than that in air, also a higher critical velocity is also obtained for filling a wide cavity than a narrow cavity, as expected. Furthermore, the filling efficiency for various powders can be distinguished from the tests with a narrow cavity, for example, larger and smoother particle powders have a higher critical velocity, which means they can fill the same cavity at higher shoe speeds. In addition, the influence of airflow and air pressure can also be estimated. It can be seen from Table 4 that the difference between the critical shoe speeds in air and vacuum is greater for fine, irregular powders. Therefore, the critical shoe velocity can be used as an index for filling efficiency.

4.3. Flow into complex shaped cavities

The kinematics of the flow process becomes more complex when the die contains features, such as steps. The details of the flow processes for filling into a stepped die are sensitive to the microscopic properties of the powder and different patterns can be observed depending on the kinematics of the shoe motion, the location of the step with respect to the direction of shoe motion, and whether the experiments are conducted in air or vacuum. A series of experiments were conducted using stepped dies of the type illustrated in Fig. 11. The top part of the cavity is equivalent to the wide cavity described above, while the introduction of the step creates a narrower section similar to the narrow cavity. As before, it is instructive to first consider the flow from a stationary shoe in vacuum.

When the powder is delivered from a stationary shoe, the initial flow is similar to that described for the wide cavity in Section 4.1. The powder flows fastest at the centre, resulting in the formation of a heap at the centre of the die on the step (Fig. 11a). Powder which falls immediately above the narrow section drops directly into it under gravity, giving a faster flow rate than for the isolated narrow cavity of Section 4.2 (the flow rate is determined by the dimensions of the wider cavity, see Eq. (1), and the particles have a finite velocity as they first enter the narrow section and are not starting from rest). As the pile formed on the step grows, it first feeds additional material into the narrow section as particles cascade over its surface (Fig. 11b), similar to the process of sand pile formation described in Section 3.2. Eventually the pile bridges across the narrow section (Fig. 11c). Powder is now fed into the narrow section by detaching from the bottom of the bridge. This closely resembles the flow process described in Section 4.1 and results in a slower flow rate into the narrow section than observed initially. For fine cohesive particles, the bridge can translate upwards as material detaches from it. Eventually the narrow section becomes completely full as the upper surface of the deposited material merges with the mass of powder in the wider channel above it (Fig. 11d).

Table 4
Critical velocities (mms^{-1}) for complete filling in air and in vacuum for various powders

Powder	Narrow cavity		Wide cavity	
	In vacuum	In air	In vacuum	In air
OL	262.36	246.28	376	360
OS	254.28	191.40	376	300
DAE	233.16	184.27	368	300
Fine sand	227.32	135.08	368	288

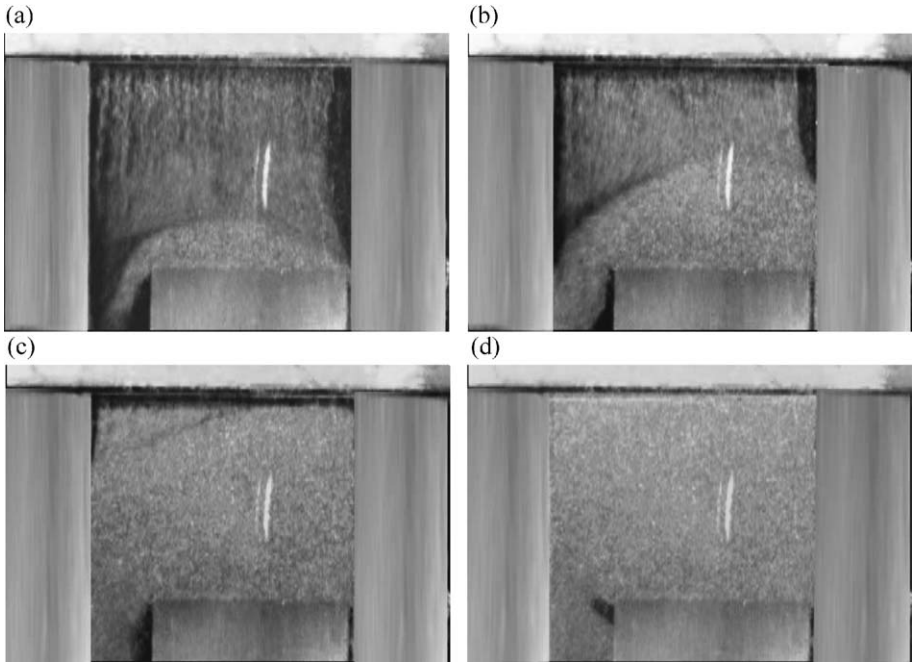


Fig. 11. Flow of DAE from a stationary shoe into a stepped die in vacuum. (a–d) The flow patterns at different instants.

The initial stages of experiments conducted in air are similar to those described above, but as soon as the pile formed on the step spreads across the narrow opening air becomes trapped in this section and the pressure rapidly builds up. For the fine particles (such as fine sand, DAE and OS), we observe two possible types of behaviour, depending on the air pressure created in the narrow section: the negative pressure acting on the bridge can either sustain the bridge longer or push the powder away to form a chimney-like channel, through which air can escape as illustrated in Fig. 12. When a chimney is formed, particles are carried up with the expelled air as powder is fed into the narrow section down the sides of the chimney (Fig. 12a). The channel formed fills from the bottom until the entire die is completely filled (Fig. 12b). In situations where a bridge is sustained, the developed pressure pushes the bridge up

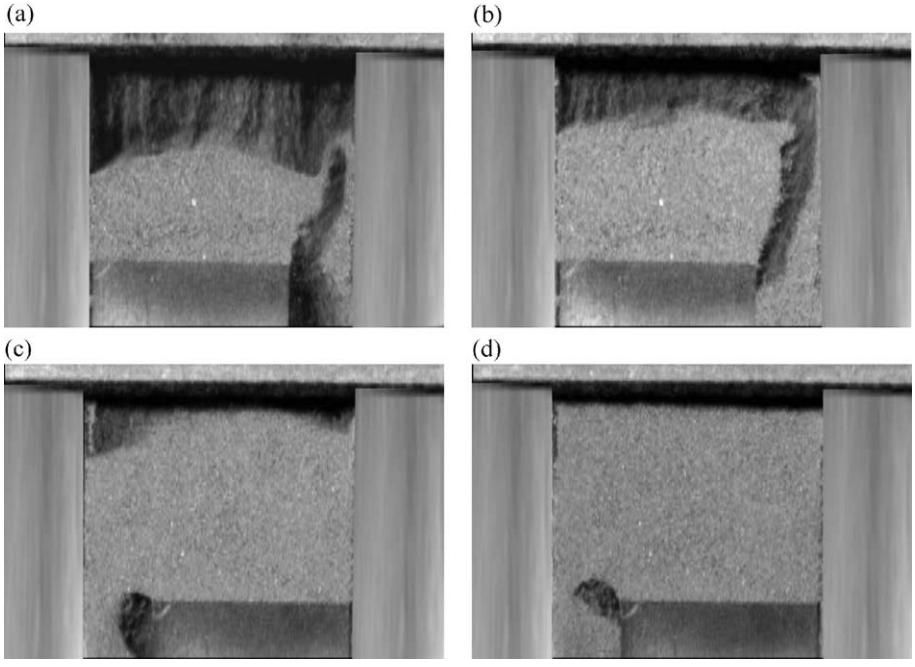


Fig. 12. Flow of DAE from a stationary shoe into a stepped die in air. (a and b) The escape of entrapped air through a chimney-like channel; (c and d) the escape of entrapped air through elevating air bubble.

until the channel below it is completely full (Fig. 12c and d). The bridge at this instant represents a surface between regions of powder which have been deposited by two different types of flow, i.e. the flows into the narrow channel and the more rapid flow into the top of the die. For coarse, heavy grains (say OL), the air can permeate through the powder stream readily. Thus, the air pressure developed in the narrow section is negligible. Consequently, the presence of the air has a very limited influence on the flow process and the flows in air are as smooth as in vacuum.

Motion of the shoe delivery system modifies the details of the flow process described above. Even for fine, light powders, how the entrapped air escapes depends on the kinematics of shoe motion and the relative orientation of the step with respect to the direction of shoe motion, because these directly determine how the air pressure builds up in the narrow section as a consequence of the different filling patterns. We consider the situation where the shoe velocity varies from zero (stationary shoe) to 400 mms^{-1}. In each of the experiments described below, the shoe moves from right to left. When a moving shoe is used, the first contact of the falling powder stream with the die gradually moves towards the leading side (left) of the die as the shoe speed is increased. Hence, different patterns are observed for flow into stepped dies with the narrow section on the right and on the left. We initially consider the situation where the narrow section is downstream of the direction of shoe motion, i.e. it is on the left (see Fig. 13a–d). At slow shoe speeds (e.g. 100 mms^{-1}), powder is deposited in the trailing side of the die as soon as the shoe starts to translate over the opening. Thus, the powder first builds up as a heap on the ledge and then cascades down the surface towards the edge of the step as the shoe moves forward, where it falls into and then bridges over the narrow section. When the bridge first forms, there is less powder in the narrow section of the die than for the stationary shoe at this stage. The subsequent response is similar to that described for the stationary shoe (Fig. 13b). In vacuum, powder is delivered smoothly into the narrow section, but for the experiments conducted in air using the fine powders, the chimney effect is stronger and more turbulent as more air needs to be evacuated from the narrow section. As the shoe speed increases further (say 200 mms^{-1}), the net flow rate of powder into the die is higher (see Figs. 8 and 9), and the intensity of the forward cascading motion over the surface of the heap into the leading corner increases. In the tests conducted in air, a chimney forms as before for the fine powders (Fig. 13c), but eventually material is deposited directly above the chimney creating a bridge towards the top of the die which is sustained by the air pressure built up beneath it. Powder is delivered gently into the void created as material detaches and falls from the surface of the bridge. At higher shoe speeds (say 400 mms^{-1}), the powder stream first flows close to the leading side with a high horizontal speed imposed by the moving shoe. Powder flows directly into the narrow section, but a heap quickly develops above the opening of the narrow section and then cascades backwards aided by the rebound of the powder stream after it hits the die wall. The air trapped in the narrow section escapes by breaking through the thinnest section of the heap at an angle of about 45° to the horizontal. As the air escapes, more powder is deposited on top of the outward flowing air and

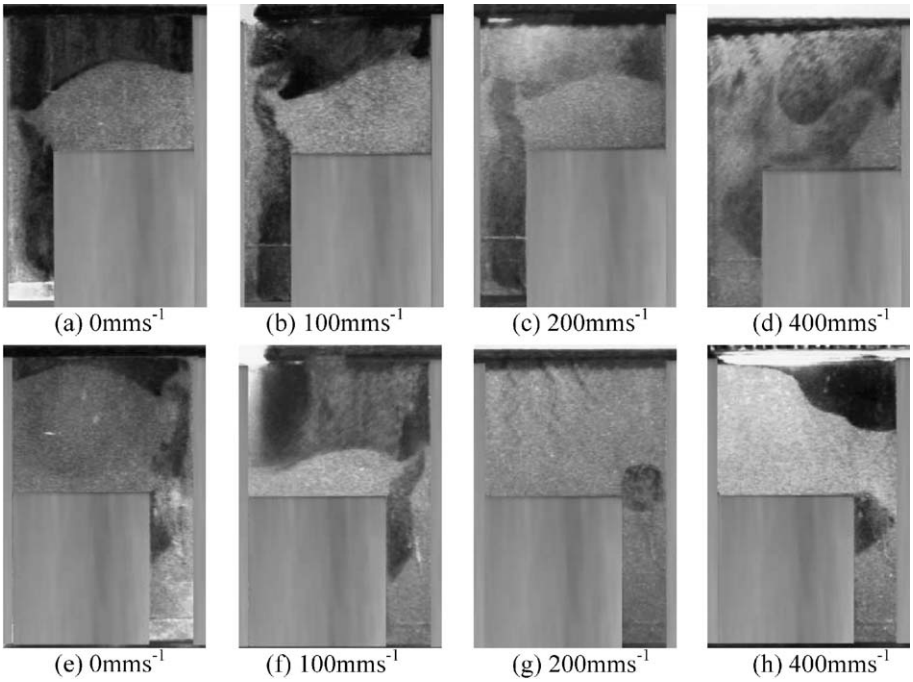


Fig. 13. Filling patterns for powder flowing into stepped dies with narrow section, (a–d) on the left and, (e–h) on the right at various shoe velocities.

significant turbulence is created in the incoming stream of powder (Fig. 13d). It is clear that the interaction between the airflow and incoming powder stream becomes stronger as the shoe speed increases, since more air is entrapped in the narrow section and stronger air pressure builds up in a short time. Flows in vacuum or in air using large particles are smoother and the filling of the narrow section is faster as particles detach more easily from the bridge formed over the narrow section.

We observe a further variation on the flow patterns when the narrow section is located on the right side of the die (see Fig. 13e–h). The flows in vacuum and for the large particles in air are smooth. Here we concentrate on the observations made for the fine particles in air. At a shoe speed of 100 mms^{-1}, powder is initially delivered directly into the narrow section as the front of the shoe moves over the die. As it progresses further, powder piles up on the ledge and flows backwards towards the edge of the step as the pile grows. Eventually the powder bridges over the narrow section. A chimney is formed as before, but there is less trapped air than when the narrow section is on the left and the chimney action is less intense (Fig. 13f). As the shoe velocity is increased, powder flows towards the leading corner of the die. A heap forms in the corner and powder particles cascade over the surface back towards the edge of the step and into the narrow section. The opening is quickly bridged. The subsequent response depends on how quickly powder piles up above the opening. At a velocity of 200 mms^{-1}, air initially breaks through the bridge and the chimney effect is observed. However, the chimney is quickly suppressed by the incoming powder stream, hence, an air bubble is formed which moves upwards (Fig. 13g) until the powder bed above it is thin enough for the air to escape by pushing the powder away. At a velocity of 400 mms^{-1}, powder builds up quickly above the narrow section and air is unable to break through to form a chimney. As the pressure builds up, the bridge translates upwards. Eventually, after the shoe has passed, the air breaks out causing a collapse of the powder that was previously above it (Fig. 13h). The escaping air carries a plume of fine particles with it. This effect is more evident for finer particles.

In each of the experiments described above, the overall rate of delivery of powder into the die is similar to and exhibits the same trends as for the filling of the wide die described in Section 4.2. Whether the narrow section is on the right or left has a very small influence on the net flow rate, but it is evident from the above discussion that the details of the filling process are very sensitive to the geometry of the die cavity. As before, for a given situation, we can determine the Beverloo constant C of Eq. (1). When using this equation, we identify A^* and D_H with the cross-sectional area and hydraulic diameter of the opening at the top of the die. The ratio of the Beverloo constant for the stepped die to that for the corresponding wide die (i.e. same material and shoe speed) for all the situations considered is in the range 0.6–0.8, with OL towards the top of this band and fine irregular particles towards the bottom. This indicates that the net rate of filling is always slower when using a stepped die. The stepped geometry is such that the die volume is less than that for the simple wide cavity. There are a number of effects which can contribute to differences in the observed values of C. There is a higher average pressure head during the filling process for the stepped die, which should result in a faster average flow rate into the die. This is counteracted by the fact that the initial stage of the flow process, during which the powder starts from rest (or at least with zero vertical velocity at time $t=0$) and accelerates towards the steady filling velocity, occupies a more significant fraction of the overall filling time. This latter effect is largely responsible for the reduction in the value of C. This further illustrates the transient nature of the filling process and that, although the Beverloo equation provides a useful context for evaluating the results of the filling experiments, C depends on the details of the filling process and only provides a qualitative interpretation of the results, i.e. care needs to be exercised in extrapolating the results beyond the conditions employed in a given set of experiments.

For filling in air, the strong effect of airflow out of the die can have a significant influence on segregation in practical powder blends, particularly if there is a significant difference in the size and/or density of the particles of the constituent materials (such as in steel powders, which often contain fine carbon particles). The effects of airflow are strongest (resulting in significant turbulence) at high shoe velocities when the narrow section is downstream of the direction of shoe motion. When the narrow section is upstream, we observe a much gentler filling process consisting of a rapid flow into the top section of the die and a gentler filling process into the lower narrow section. The density variations within the die that can result from these different types of flow are examined more fully elsewhere [21] where they are compared with the predictions of computer calculations using discrete element methods (see also Ref. [22]). As might be expected, a lower density is observed in the lower narrow section, and the merge plane formed between the different flows provides a potential region of low density.

5. Discussion

The experiments described above demonstrate the influence of shoe kinematics, die and shoe geometry and airflow on the details of the filling process and the way in which the powder packs within the die. Two types of flow have been identified, which we refer to as nose flow and bulk flow. Nose flow involves particles cascading down the slope of a nose-shaped profile created by inertia, frictional interaction of the powder with the surface on which the shoe slides and the initial flow of powder into the die. Nose flow is not strongly influenced by the presence of air in the cavity. The gap created at the front of the cavity allows rapid air evacuation as powder falls into the die. Also, the cascading

and detachment processes are not sensitive to the presence of air. Bulk flow on the other hand is strongly influenced by the presence of air and significant differences are observed between experiments conducted in air and vacuum. The air present during die filling has four major effects on the flow process: (1) the expansion of the powder as it falls from the shoe into the die creates an adverse pressure gradient which opposes the motion of the particles; (2) pressure built up by air trapped in the die also opposes the flow of powder into the die; (3) air flowing between the particles provides a lubricating effect, allowing the particles to effectively slide past each other more easily; (4) finally, the drag force exerted by the air as it escapes from the die also slows down the flow process—it can in fact result in particles, particularly if they are light, being carried with the upward flow of air and either being ejected from the die or interacting with the downward particle flow resulting in turbulence and circulation of the powder mass. The development of efficient filling procedures requires all these effects to be taken into account.

It is evident from the above discussion that nose flow is an inherently faster filling process than bulk flow. This observation is reinforced by the results from a series of experiments in which the volume of powder placed in the shoe was varied. In the experiments reported here, about 75% of the available shoe volume was filled with powder. Completely filling the shoe hinders the creation of a nose profile and reduces the contribution of nose flow to the filling process, resulting in a decrease in the net flow rate of powder into the die. This beneficial effect of nose flow must be balanced by the fact that at low shoe velocities, only a small portion of the available cross-sectional area of the die opening is used when the die is filled. Under these conditions, the net flow rate can be increased by creating multiple noses. This can be achieved by dividing the shoe into a number of compartments using transversal separating walls, creating a series of powder profiles similar to that shown in Fig. 6. Experiments were conducted using similar kinematic conditions to that described in Section 4 for the flow of metal powders (OL, OS and DAE) into a wide cavity. It was found that shoes containing two or three cells lead to higher filling ratios over a range of shoe speeds, increasing the critical velocity below which there is complete filling compared with that for a single compartment. It is found that the improvement of die filling ratio becomes more significant for finer and more irregular materials (DAE). Increasing the number of compartments further hinders the formation of nose type profiles. Introduction of a large number of compartments creates long narrow columns of powder. Nose type profiles are then unable to form. In addition, friction along the dividing walls hinders the flow of powder into the die, with the delivery being equivalent to the mass flow described for hoppers in Section 3.1, which is a slower delivery process than core flow, which is the mechanism responsible for bulk flow described above [23].

Nose flow does not suffer from any of the detrimental effects due to the presence of air described above. At an elementary level, it can be thought of as creating an easy channel for air to escape. This prevents the build up of pressure in the die and results in a smoother air evacuation process, which limits the effect of drag on the falling particles. Clearly, this type of situation can be promoted in other ways, by, for example, creating channels in the shoe for easy air evacuation [23]. This is commonly employed in practical filling systems, but this type of set-up does not benefit from the other advantageous effects of nose flow, such as the freer cascading motion down a free surface and absence of pressure gradients in the expanding powder mass.

The improvements described above are only realisable in simple dies. For the flow into a complex cavity as discussed in Section 4.3, shoe design has a limited effect on the major characteristics of the filling process in the vicinity of any discontinuities in the die geometry. Air can become trapped in narrow sections and this has a strong influence on the way in which powder is deposited in these regions. The nature of the filling process depends on the location of these regions with respect to the direction of shoe motion. In narrow sections where air is trapped, we observe a gentler filling process than in the rest of the cavity. How this influences the filling density is examined more fully elsewhere [21].

In this paper, we have concentrated on the process of gravity filling. Different flow patterns and density distributions are likely to occur in other methods, such as suction filling or fluidisation methods. Filling using an aeration and/or fluidisation shoe has recently been developed to improve the filling process [24,25]. In this system, gases of low pressure are injected into the shoe. Although it is still unclear how this system works, it has been shown that this technique can increase the flowability of powder and the consistency of compact PM parts of certain geometry. It is believed that this technique appears to promote the third effect described above, i.e. that the injected gas lubricates the powder. In addition, the fluidisation process probably opens up the channels between the particles as they flow into the die allowing the air to escape more easily. Hence, it may promote faster air evacuation. Further studies are, however, required in order to more fully evaluate the merits and suitability of fluidisation techniques.

The Beverloo equation presented in Section 3.1 provides a useful starting point for the design of shoe filling systems. It can be used to determine the maximum speed at which the cavity is completely filled and to determine the time to fill intricate sections of the die compared with that required to fill the bulk of the cavity.

6. Conclusions

Experimental studies of powder flow into constrained cavities of simple and complex geometries have been presented. Die filling tests have been carried out for various powders either in air or in vacuum. The filling processes were recorded using a high-speed video system. Both qualitative

and quantitative descriptions of the flow process have been given. It has been shown that the powder characteristics, shoe speed, die geometry and airflow play important roles in the filling process. The filling of powder into a cavity can be classified into nose flow, which dominates the filling process at low shoe speed, and bulk flow which dominates at high shoe speed. It is found that the filling rate decreases as irregularity of particle morphology increases and as the average particle size decreases. The shoe speed also significantly affects the filling process. A lower filling rate is obtained at high shoe speeds.

The presence of air during die filling has a significant influence on the filling process. The disadvantages of air during die filling are (1) the pressure built up by air trapped in the die opposes the flow of powder into the die; and (2) the drag force exerted by the air as it escapes from the die slows down the flow process and it can also carry fine particles with it. Comparisons of experimental results in air and in vacuum reveal that the filling rate is much slower in air than in vacuum. Therefore, promoting faster air evacuation could benefit the practical filling process. The advantage of air in die filling is that air flowing between the particles provides a lubricating effect, allowing the particles to effectively slide past each other more easily. This can increase the flowability of the powder, as illustrated by Kohdon et al. [24] and Zahrah and Rowland [25]. Further work is necessary, however, to explore whether this lubricating effect can lead to uniform and higher density distributions in a constrained cavity.

Even though the Beverloo equation was originally proposed for steady-state flows and it is not applicable for the transient cases considered in this paper, an interpretation of the die filling process in the context of the Beverloo equation provides a useful framework for the design of shoe filling systems. Examination of how the Beverloo constant is influenced by the powder characteristics, air flow and pressure, and shoe speed has led to the concept of a critical velocity as providing a measure of the flowability of a powder during die filling.

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References

[1] R.M. Nedderman, Statics and Kinematics of Granular Materials, Cambridge Univ. Press, New York, 1992.
[2] J.P.K. Seville, U. Tuzun, R. Clift, Processing of Particulate Solids, Blackie Academic and Professional, London, 1997.
[3] S.B. Savage, The mechanics of rapid granular flows, Advances in Applied Mechanics 24 (1984) 289–366.
[4] R. Greeve, T. Koch, K. Hutter, Unconfined flow of granular avalanches along a partly curved surface: I. Theory, Proc. R. Soc. Lond., A 445 (1994) 399–413.
[5] L.R. Lawrence, J.K. Beddow, Powder segregation during die filling, Powder Technology 2 (1968/1969) 253–259.
[6] L.R. Lawrence, J.K. Beddow, Some effects of vibration upon powder segregation during die filling, Powder Technology 2 (1968/1969) 125–130.
[7] C. Demetry, F.S. Souto, B.C. Rydaen, J.M. Roy, Tactile sensing of density uniformity in powder beds after die filling, Powder Technology 99 (1998) 119–124.
[8] G.F. Bocchini, Influence of small die width on filling and compacting densities, Powder Metallurgy 30 (1987) 261–266.
[9] E.R. Rice, J. Tengzelius, Die filling characteristics of metal powders, Powder Metallurgy 29 (1986) 183–194.
[10] J.A.S. Cleaver, R.M. Nedderman, Theoretical prediction of stress and velocity profiles in conical hoppers, Chemical Engineering Science 21 (1993) 3693–3702.
[11] U. Tuzun, G.T. Houlsby, R.M. Nedderman, S.B. Savage, Flow of granular materials: 2. Velocity distributions in slow flow, Chemical Engineering Science 37 (1982) 1691–1709.
[12] W.A. Beverloo, H.A. Leniger, J. van de Velde, The flow of granular solids through orifices, Chemical Engineering Science 15 (1961) 260–269.
[13] R.M. Nedderman, U. Tuzun, S.B. Savage, G.T. Houlsby, Flow of granular materials: 1. Discharge rates from hoppers, Chemical Engineering Science 37 (1982) 1597–1609.
[14] S.B. Savage, R.M. Nedderman, U. Tuzun, G.T. Houlsby, Flow of granular materials: 3. Rapid shear flows, Chemical Engineering Science 38 (1983) 189–195.
[15] B.J. Crewdson, A.L. Ormond, R.M. Nedderman, Air-impeded discharge of fine particles from a hopper, Powder Technology 16 (1977) 197–207.
[16] T.M. Verghese, R.M. Nedderman, The discharge of fine sands from conical hoppers, Chemical Engineering Science 50 (1995) 3143–3153.
[17] S. Jing, H. Li, Study on the flow of fine powders from hoppers connected to a moving-bed standpipe with negative pressure gradient, Powder Technology 101 (1999) 266–278.
[18] G. Ferrari, M. Poletto, The particle velocity field inside a two-dimensional aerated hopper, Powder Technology 123 (2002) 242–253.
[19] O. Narayan, S.R. Nagel, Incipient failure in sandpile models, Physica A 264 (1999) 75–83.
[20] J.P. Wittmer, P. Claudin, M.E. Cates, J.-P. Bouchaud, An explanation for the central stress minimum in sans pile, Nature (Letters) 382 (1996) 336–338.
[21] C.Y. Wu, A.C.F. Cocks, L. Dihoru, Experimental and numerical investigation of filling density variation, Powder Metallurgy, 2003 (to be submitted).
[22] C.Y. Wu, A.C.F. Cocks, O.T. Gillia, Experimental and numerical investigations of die filling and powder transfer, Advances in Powder Metallurgy and Particulate Materials, Part 4 (2002) 258–272.
[23] A.C.F. Cocks, L. Dihoru, T. Lawrence, A fundamental study of die filling, Proceedings of PM2001 (2001) 255–260 (Nice, France).
[24] M. Kondoh, S. Takemoto, I. Urata, Development of uniform powder filling method-aeration powder filling, Proceedings of the 1998 PM World Congress-Granada, Spain (1998) 85–90.
[25] T.F. Zahrah, R. Rowland, Use of fluidisation for die filling applications, P/M Science and Technology Briefs 4 (2002) 24–27.